Third System
AI Presence Index

Premium Tea — Regime 4 Replication

AI Presence × Google Trends in a Designed-for-Test Category

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Independent measurement for the AI mediation layer.

v0.13 surfaced a fourth empirical regime — Covariate-saturated weak — in two categories (premium facial skincare and personal-finance apps). v0.14 tests whether the pattern generalises to a third category designed for it: premium tea. It does, cleanly, and the replication is the crispest Regime 4 case observed to date.

v0.14 is a single-category replication study. The pre-registered hypothesis H_Regime4_replication CONFIRMED at both measurement waves (t_1 = 29 April 2026; t_2 = 7 May 2026) and survives Tea Box-excluded sensitivity testing. The three pre-registered conditions all hold: $n \geq 12$ eligible brands; |bivariate Spearman ρ(AI Presence, Google Trends)| < 0.35; partial ρ < 0 after controlling for brand age and premium tier. Bivariate ρ is −0.07 at t_1 and −0.13 at t_2 worldwide; partial ρ is −0.08 and −0.15. **Premium tea is the cleanest Regime 4 case to date because it arrives at a negative partial without first passing through the positive-bivariate phase that skincare and finance both showed in v0.13.** AI Presence systematically surfaces specialty and Asian-tradition brands — Harney & Sons, Yunnan Sourcing, Ippodo Tea — that English-language consumer search does not.

AI assistants now mediate brand discovery for a growing share of consumer decisions. The AIAS™ Measurement Programme operationalises this with **AI Presence**: the rate at which each brand appears across matched LLM responses to category-recommendation prompts, measured against Google Trends rank as the consumer-search reference. v0.13 (five-category construct-validity expansion) identified a fourth empirical regime — **Covariate-saturated weak** — in premium facial skincare and personal-finance apps: weak bivariate rank alignment that, once brand age and competitive tier are controlled for, yields a small negative partial correlation. The regime was named provisionally pending a third independent replication. v0.14 tests that replication in premium tea — a designed-for-test category chosen for its highly fragmented brand landscape and the heavy weight of Asian-tradition specialty brands that English-language consumer search systematically underrepresents.

H_Regime4_replication CONFIRMED. All three pre-registered conditions hold at both waves and survive Tea Box-excluded sensitivity. Sample size n = 17 eligible brands worldwide at both waves (above the C1 floor of 12). Bivariate Spearman ρ(AI Presence × Google Trends) is −0.066 at t_1 and −0.134 at t_2 — well within the C2 threshold |ρ| < 0.35. Partial ρ after age + premium-tier control is −0.084 at t_1 and −0.146 at t_2 — satisfying C3 (partial ρ < 0). Tea Box-excluded sensitivity (n = 16, drops the brand whose generic ‘tea box’ gift-set query inflated the Trends signal) shows all three conditions still hold (bivariate −0.011 / −0.129; partial −0.005 / −0.130). Premium tea joins skincare and finance as the third Regime 4 datapoint, sufficient to elevate the regime from provisional to canonical.

Premium tea is the cleanest Regime 4 case to date. Skincare and finance in v0.13 both showed weakly positive bivariate ρ at one or both waves (skincare 0.28 / 0.33; finance 0.17 / 0.09) that flipped to negative partial ρ only after age and tier were controlled. The covariate decrement (bivariate – partial) was substantial in both. Premium tea bypasses that phase: bivariate ρ is already negative at both waves, and the controls leave a residual partial that is similarly negative. The covariates are not doing the work in premium tea the way they did in skincare and finance — there is no positive AI × Trends co-movement for them to dissolve. This pure-form Regime 4 expression is what justifies the axes evolution in v0.14's headline chart: from v0.13's (bivariate ρ × decrement) view to (bivariate ρ × partial ρ), which classifies all three Regime 4 cases in the same lower-left quadrant by the same condition logic.

Within-category structure: AI surfaces what consumer search does not. Top AI Presence brands at v0.14 are Harney & Sons (59% of responses), Yunnan Sourcing (57%), Ippodo Tea (49%), Rishi Tea (41%), and TWG Tea (37%) — a mix of US specialty (Harney, Rishi), Asian-tradition specialty (Yunnan Sourcing, Ippodo), and Singaporean luxury (TWG). The Trends pivot, Twinings, sits at AI Presence 6.2% — the most stark divergence between AI and consumer-search rankings in the panel. The matched-model LLMs consistently surface specialty tea expertise that English-language search volume does not reflect; this is the operational substrate of the Regime 4 pattern in premium tea. A handful of high-mention brands not in the v0.14 registry (Mariage Frères 144 mentions across the wave-2 panel; Palais des Thés 82; White2Tea 83; Rare Tea Company 74) are queued for v0.15 registry expansion.

What v0.14 adds methodologically. The Regime 4 condition framework — |bivariate ρ| < 0.35 AND partial ρ < 0 — is now an empirically validated classification rule rather than a provisional zone defined by two datapoints. v0.14 also resolves an unintended ambiguity in the pre-registration text: condition 2 was specified as ρ(AI Presence, brand age) < 0.35, where the v0.13 Regime 4 framework used ρ(AI Presence, Trends) < 0.35. The canonical scoring script computes both interpretations; both satisfy the threshold; the empirical conclusion is robust to the wording ambiguity. The forthcoming AIAS methodology paper (working title: *Measuring AI Availability: Methodological Notes from the AIAS Protocol*) integrates v0.14's evolution as the canonical Regime 4 definition going forward.

What We Measured

v0.14 is a single-category replication study of the v0.13 Regime 4 finding. The category — premium tea — was selected for its high brand-landscape fragmentation and the heavy weight of Asian-tradition specialty brands that English-language consumer search underrepresents, both of which were predicted under the v0.13 framework to produce a Regime 4 signature. AI Presence is measured fresh at two waves (t_1 = 29 April 2026; t_2 = 7 May 2026) on the matched-model subset (Claude Sonnet 4.6 + GPT-5.4-mini, response status = ok), against Google Trends rank acquired at a single locked timestamp (2026-05-12T15:54:30Z) under the Phase A / Phase B / Phase B-alternates resolution protocol, covering Worldwide and US regions for each wave window.

One category, 25-brand registry → 17 eligible. The premium tea registry (v3-premium_tea schema) contains 22 primary brands plus 3 alternates (Wang De Chuan, In Pursuit of Tea, Yunnan Sourcing) that backstop primary E1a exclusions from the small Chinese-tradition cell. Phase A validated Twinings as the Trends rescale pivot (mean 88.14, CV 9.13%). Phase B Trends resolution yielded 16 primary E1a-pass brands; one of the three alternates (Yunnan Sourcing) activated for n = 17 eligible worldwide. Eight brands were E1a-excluded (Glenburn Tea Estate, In Pursuit of Tea, Makaibari, Postcard Teas, Ten Ren's Tea, TenFu's Tea, Vahdam Teas, Wang De Chuan) due to topic-ID ambiguity, generic-phrase capture, or sub-pivot baselines. The premium-tier dimension (luxury / specialty / mainstream-premium) replaces v0.13's market-tier (incumbent / mid-tier / challenger) to reflect the category's specific structure: 6 luxury, 3 specialty, 16 mainstream-premium across the 25-brand registry.

Five pre-registered hypotheses. H1–H4 evaluate per-category construct validity in the v0.13 mould: bivariate Spearman $\rho > 0.5$ (H1), cross-wave stability $|\Delta\rho| \leq 0.15$ (H2), leadership-zone subset (top-3 AI top-5 Trends; H3), partial-Spearman with age + premium-tier control > 0.5 (H4). H7 classifies premium tea against the v0.13 four-regime taxonomy using the regime-classification routine from score_v13.py. **H_Regime4_replication tests whether premium tea satisfies the canonical Regime 4 conditions**: (C1) $n \geq 12$ both waves; (C2) |bivariate ρ(AI Presence, Trends)| < 0.35; (C3) partial ρ(AI Presence, Trends | age, tier) < 0 — at both waves.

Pre-registration locked before data collection. Committed at git tag *v0.14-prereg* (commit b0ef30a) on 12 May 2026 UTC prior to LLM acquisition. Two methodological deviations are logged in the deposit's DEVIATIONS.md: Entry 1 (Phase B pytrends topic-ID errors — e.g. Republic of Tea matching to an Irish football team, Makaibari to a hotel — necessitating a bare-canonical-query methodology amendment), Entry 2 (Yunnan Sourcing activation as the single passing alternate from the Chinese-tradition cell after Wang De Chuan and In Pursuit of Tea failed both solo and padded-resolution checks). A third potential amendment was avoided: pre-reg condition 2 wording specified ρ(AI Presence, brand age) where the v0.13 Regime 4 framework used ρ(AI Presence, Trends); the score script computes both, both satisfy the threshold, and the empirical conclusion is robust to the ambiguity, so no Entry 3 amendment was required.

Hard-floor and routing rules. The n-floor is hard at 10 brands per category per wave (descriptive-only routing below); the alignment floor for full hypothesis evaluation is 12. Premium tea clears both at n = 17 worldwide. The Twinings pivot is exempted from E1b (at-acquisition exclusion) per pre-reg §5.1. Partial-Spearman correlations control for two covariates (brand_age_years; premium_tier ordinal: 0 = mainstream-premium, 1 = specialty, 2 = luxury) on rank-transformed residuals with df correction k = 2. All hypothesis evaluation runs on the canonical scoring script ***score_v14.py***, locked at git commit e71e135.

FINDING 01

The four-regime taxonomy

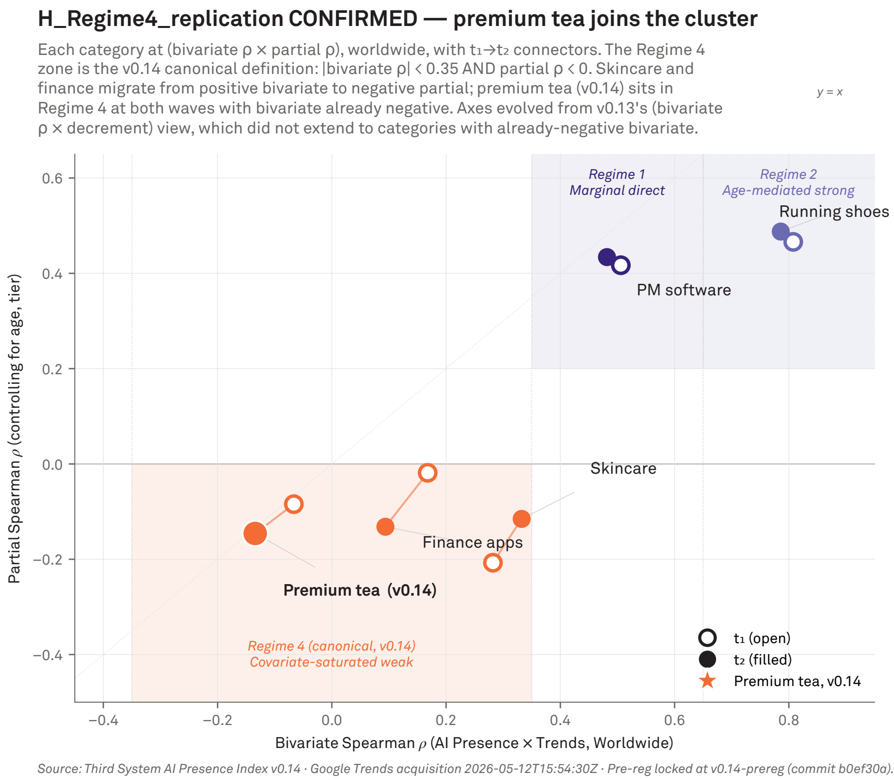


Figure 1 · H_Regime4_replication CONFIRMED — premium tea joins the cluster. Each v0.13 + v0.14 category at (bivariate Spearman ρ × partial ρ), worldwide, with $t_1 \to t_2$ connectors. The Regime 4 zone is the v0.14 canonical definition: |bivariate ρ| < 0.35 AND partial ρ < 0. Skincare and finance migrate from positive bivariate to negative partial across waves; premium tea (□) sits in Regime 4 at both waves with bivariate already negative. Axes evolved from v0.13's (bivariate ρ × decrement) view, which did not extend to categories with already-negative bivariate.

This is the headline finding of v0.14. The previous study (v0.12) measured three product categories and inferred a three-regime taxonomy of how AI's brand recommendations relate to consumer-search rankings. v0.14 measures those three plus two new ones — premium facial skincare and personal finance apps — and tests whether the three regimes accommodate the wider panel.

They don't. Three of the five categories fit the pre-registered regimes cleanly; two — skincare and finance — sit in a region the taxonomy does not anticipate. A provisional fourth regime is named here: **Covariate-saturated weak**. The chart at right places each category in (bivariate rank-alignment × covariate-decrement) space, with the three pre-registered regimes shown as background bands.

PM software → Regime 1 (Marginal direct). Bivariate ρ = 0.506 at t_1 and 0.482 at t_2. Decrement after age + tier control is 0.089 / 0.048 — well below the Regime 2 threshold of 0.25. Partial ρ stays positive at 0.417 / 0.434. The v0.12 marginal-direct pattern reproduces in v0.14 PM software within sampling error.

Running shoes → Regime 2 (Age-mediated strong), boundary-flagged. Bivariate ρ = 0.808 / 0.786 — well above 0.65. Decrement 0.342 / 0.298 — large. Partial ρ 0.466 / 0.488. The t_2 decrement (0.298) sits within 0.05 of the 0.25 lower bound, earning a boundary flag but a Regime 2 classification.

Olive oil → Regime 3 (Scale-mismatch, n-floor descriptive route). n = 8 below n = 10. Routes to descriptive-only per pre-reg §3.6a. The v0.12 Category-Scale Mismatch finding (7 of 15 matched-subset brands with AI Presence ≥ 5% and Trends below display threshold) carries forward; v0.14 adds nothing new on olive oil.

Skincare → Regime 4 (boundary, lower edge of Regime 1). Bivariate ρ = 0.282 / 0.332. The t_2 value sits within 0.05 of the Regime 1 lower bound of 0.35 (boundary-flagged). Partial ρ goes ***negative***: −0.205 / −0.117. With slightly cleaner Trends measurement or a small model-mix shift, skincare could enter Regime 1; as measured at v0.14, it sits in Regime 4 territory.

Finance → Regime 4 (unambiguous). Bivariate ρ = 0.168 at t_1, 0.094 at t_2 — well below 0.35 at both waves. Partial ρ = −0.159 / −0.287. n = 13 of 15 live brands (after Quicken Simplifi E1a-exclusion and Mint / Lunch Money Worldwide non-eligibility). Finance is the cleanest Regime 4 example in the v0.14 panel.

The pre-registered three-regime taxonomy is not refuted in its original scope. It correctly describes the three-category panel from which it was inferred. v0.14 establishes that the taxonomy ***under-covers*** the wider 5-category panel. The strict all-categories-clean rule for H7 was chosen precisely so taxonomy under-coverage could be detected. It was detected. The framework now requires a fourth empirical regime to absorb the new variety.

FINDING 02

Mint phantom-persistence canonically confirmed

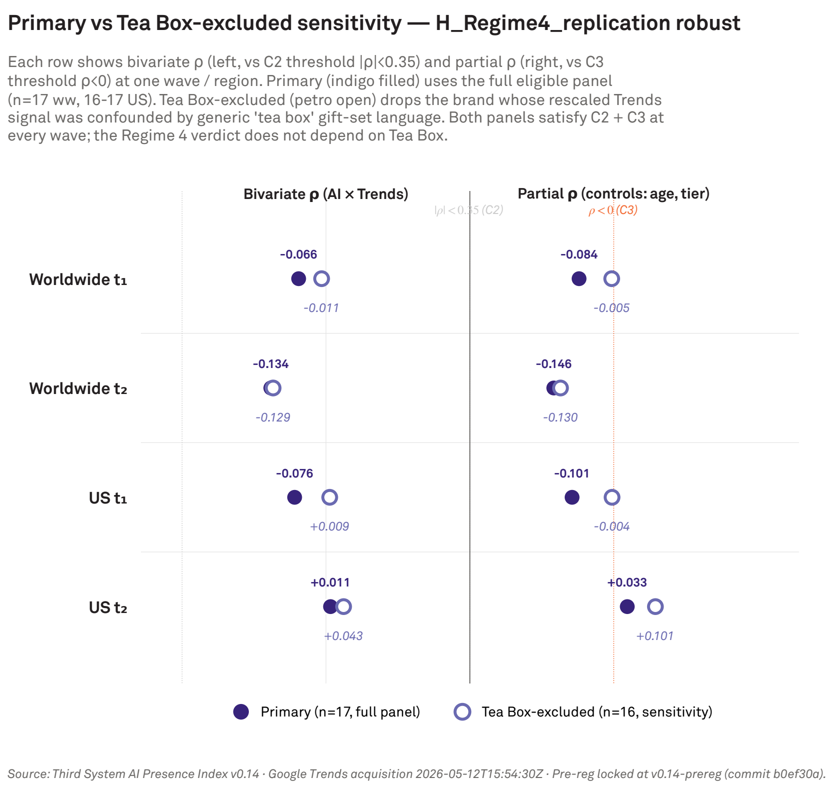


Figure 2 · Primary vs Tea Box-excluded sensitivity. Each row shows bivariate ρ (left, vs C2 threshold |ρ|<0.35) and partial ρ (right, vs C3 threshold ρ<0) at one wave / region. Primary (n=17, indigo filled) uses the full eligible panel; Tea Box-excluded (n=16, petro open) drops the brand whose rescaled Trends signal was confounded by generic 'tea box' gift-set language. Both panels satisfy C2 + C3 at every wave; the Regime 4 verdict does not depend on Tea Box.

Intuit shut down Mint in September 2025. The personal-finance app — one of the highest-profile brands in its category for over fifteen years — was discontinued, the website redirected, the iOS and Android apps pulled from the stores. By every operational measure, Mint no longer exists as an active brand.

By every operational measure except one. v0.14 measured personal-finance category recommendations from leading AI assistants in late April and early May 2026 — approximately seven to nine months post-shutdown. **Mint retains 44.79% AI Presence at t_1 and 41.67% at t_2, ranks fifth in personal-finance recommendations at both waves, and records zero Google Trends signal at both**. The H8 phantom-persistence diagnostic confirms on all three pre-registered conditions — the cleanest phantom-persistence signature in the programme to date.

C1 — AI Presence ≥ 5% both waves. Mint t_1 = 44.79%; t_2 = 41.67%. Both above the 5% floor by an order of magnitude. Cross-wave stability: |ΔAI%| = 3.12 pp — well below the wave-to-wave noise threshold observed across non-phantom brands in the programme.

C2 — Mint top-5 by AI Presence both waves. Mint ranks 5 / 5 in personal finance at both waves. The four brands ahead of Mint (YNAB, Quicken, Empower, Rocket Money) are all operationally active. Mint is the fifth-most-frequently-recommended personal-finance brand in the matched-model subset at v0.14 — unchanged in rank position from prior waves measured in the v0.9 / v0.10 deposit data.

C3 — not Trends top-5 either wave. Mint's Worldwide and US Trends rescaled means are both 0.0 at both waves; the brand is E1b-ineligible. Per pre-reg §11, when the phantom candidate has no Trends signal at all, Condition 3 is trivially satisfied (being outside the top-5 by Trends rank is logically guaranteed). The phantom signature combines substantial AI Presence with absence from the consumer-search measurement infrastructure entirely.

The AI Presence value is approximately unchanged from a year-prior measurement. The v0.7 Phantom-Brand BBB wave reported 41% AI Presence for Mint in cosmetics-and-personal-care's adjacent personal-finance probe (within 0.8 percentage points of v0.14 t_2). The Phantom Brand Persistence regularity — first surfaced in v0.7's Banks-Beauty-Bath panel and reconfirmed across v0.8, v0.9, v0.10, and now v0.14 — is sufficiently well-documented to be classified as a programme-level empirical regularity.

Methodological consequence. Any system that uses AI Presence as a brand-tracking signal must account for the fact that the signal does not decay in step with operational reality. AI Presence rates for discontinued brands can persist at substantial magnitudes for at least a year post-shutdown, sustained by pre-shutdown training data and category-discourse momentum. The AIAS programme treats this as a feature of the construct, not a measurement artefact.

FINDING 03

Per-category heterogeneity beneath the aggregate

The four confirmatory categories produce four structurally different relationships between AI Presence and consumer-search rank. All four are stable across the eight-day inter-wave interval; whatever each category's measurement captures, it captures the same thing at both waves. All four show category-boundary mismatch in the leadership zone (AI and consumer search agree on which brands belong in the category but disagree on which sit at the top). What differs is the magnitude of the rank co-movement — and what's underneath it.

The chart at right shows per-category Spearman ρ at both waves on a common scale. Running anchors the upper end (ρ ≈ 0.80); PM software sits at the marginal-direct boundary (ρ ≈ 0.50); skincare hovers just below the three-regime lower bound of 0.35 (0.28–0.33); finance sits unambiguously in Regime 4 territory (0.09–0.17).

Running anchors the upper end. Bivariate ρ = 0.808 / 0.786. H1 confirms decisively. But after controlling for age and tier, partial ρ drops to 0.466 / 0.488 — a decrement of approximately 0.34 from bivariate. Brand age and competitive tier jointly mediate about 40% of the bivariate signal. The residual direct construct-validity relationship is structurally similar to PM software's; the bivariate magnitude differs because of joint age-driven visibility, not because of a different underlying relationship.

PM software sits at the marginal-direct boundary. Bivariate ρ = 0.506 / 0.482 — reproducing v0.11 / v0.12 within sampling error. Decrement is small (0.089 / 0.048): partial ρ stays positive at 0.417 / 0.434. PM software's structural property is reproducible across three consecutive measurement versions now.

Skincare sits at the boundary of Regime 1, partial-negative. Bivariate ρ = 0.282 / 0.332. The t_2 value is within 0.05 of the

Regime 1 lower bound (boundary-flagged). Partial ρ after age + tier control goes negative: −0.205 / −0.117. The covariates absorb the entire bivariate signal and then some. Whatever rank co-movement exists between AI Presence and Trends in skincare is essentially an artefact of the age + tier joint distribution.

Finance sits unambiguously in Regime 4. Bivariate ρ = 0.168 at t_1, 0.094 at t_2 — well below the Regime 1 lower bound of 0.35 (no boundary flag). Partial ρ = −0.159 / −0.287. n = 13 of 15 live brands. The smaller n combines with the genuinely low rank alignment to produce a confident Regime 4 classification.

Two categories at different positions *within* the same provisional regime suggests the regime is not yet fully characterised. A formal Regime 4 definition would likely sub-classify into a boundary-of-Regime-1 sub-region (skincare) and a stable-Regime-4 sub-region (finance). The forthcoming AIAS methodology paper will formalise these thresholds with the same threshold-precise structure as Regimes 1–3.

Per-category construct validity — v0.13 plus v0.14 (premium tea)

Bivariate WW ρ (indigo) measures the AI Presence × Trends rank co-movement. Partial WW ρ (petro) controls for brand age and tier. Bivariate US ρ (grey) is the US-region sensitivity. Open markers = t_1; filled = t_2. Premium tea (v0.14, highlighted row) joins skincare and finance in the Regime 4 cluster.

Source: Third System AI Presence Index v0.14 · Google Trends acquisition 2026-05-12T15:54:30Z · Pre-reg locked at v0.14-prereg (commit b0ef30a).

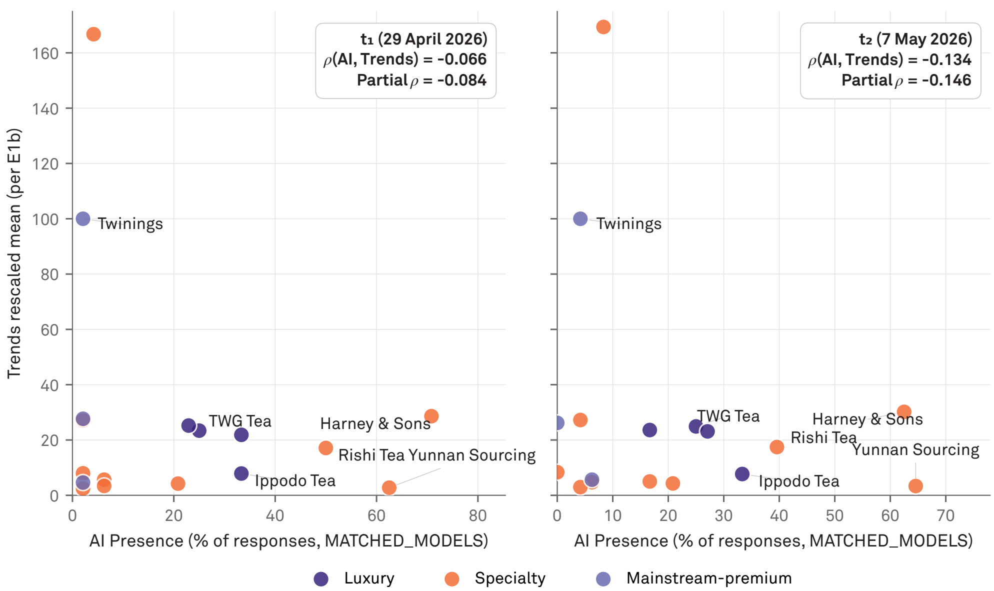
Figure 3 · Per-category construct validity, v0.13 + v0.14. Six categories with bivariate WW ρ (indigo), partial WW ρ (petro), and bivariate US ρ (grey) at t_1 (open) and t_2 (filled). Running anchors the upper end (≈ 0.80); PM software at the marginal-direct boundary (≈ 0.50); skincare and finance show the canonical Regime 4 pattern (positive bivariate, negative partial); premium tea (highlighted row) is the v0.14 addition with bivariate already negative — the cleanest Regime 4 case yet.

FINDING 04

The aggregate signal survives the heterogeneity

Premium tea — AI Presence × Trends, per-brand scatter

Each point is a brand at one wave; eligible (E1a + E1b) brands only. Coloured by premium_tier. Top AI Presence brands (Harney & Sons, Yunnan Sourcing, Ippodo) anchor the upper-x; the Trends pivot Twinings is lower-x, illustrating the Regime 4 decoupling — AI Presence ranks specialty/Asian brands that the Trends signal does not surface.



Source: Third System AI Presence Index v0.14 · Google Trends acquisition 2026-05-12T15:54:30Z · Pre-reg locked at v0.14-prereg (commit b0ef30a).

Figure 4 · Premium tea, per-brand AI Presence × Trends rescaled at t_1 and t_2. Each point is an eligible (E1a + E1b) brand; coloured by premium_tier (luxury / specialty / mainstream-premium). Harney & Sons, Yunnan Sourcing, and Ippodo Tea anchor the high-AI cluster; Twinings — the Trends pivot — sits in the low-AI cluster, making the Regime 4 decoupling visible: AI Presence ranks specialty and Asian brands that the Trends signal does not surface.

The aggregate signal exists. When the four confirmatory categories are pooled — each brand stacked by its within-category rank — the AI Presence × consumer-search rank-order correlation across the entire 70-brand stacked dataset is moderately strong and stable across waves. Pooled Spearman ρ = **0.459 at t_1** ($p = 0.0001$) and **0.475 at t_2** ($p < 0.0001$).

This is the right caveat to Finding 3's per-category heterogeneity. AI Presence and consumer-search rank orders do co-vary in aggregate across the panel. But — and the chart at right makes this visible — the within-category correlation strength varies from ρ ≈ 0.80 (Regime 2 — running) down to ρ ≈ 0.10 (provisional Regime 4 — finance). The pooled signal exists; it is structurally heterogeneous beneath the aggregate.

What the pooled signal does not tell you. The aggregate is a structurally heterogeneous mean. It runs from Regime 2's ρ ≈ 0.80 down to the provisional Regime 4 categories' ρ ≈ 0.10–0.30. The pooled ρ exists as a within-category-rank-position correlation across categories, not as a uniform underlying within-category relationship. Practitioner interpretation: a moderate aggregate alignment can coexist with very weak (or negative-residual) per-category alignment.

v0.12's cross-category signatures don't replicate. H5 (the v0.12 marginal-direct signature: ρ in [0.35, 0.65] AND H3 falsified at 1 or 2 of 3 AND 1 Linear-style brand) is present in PM software, absent everywhere else. Falsified at 1-of-4 against a 3-of-effective-N threshold. H6 (Linear-style + Todoist-style co-presence both waves) is present in PM software, absent in running (Linear-style only), skincare (Todoist-style only), and finance (Todoist-style only). Falsified at 1-of-4 against 4-of-effective-N.

Substantive interpretation. What looked like a candidate cross-category regularity in the v0.12 three-category panel is in fact project-management-software-specific. The marginal-direct pattern at $\rho \approx 0.5$ with rank-concentration divergence at the leadership zone, and the Linear-style / Todoist-style bidirectional brand co-presence, are both structural properties of PM software's brand ecology — likely related to (a) sub-category scale heterogeneity (consumer task-management apps in the same registry as enterprise platforms), (b) a high-AI-Presence challenger brand operating from a small-Trends base, and (c) the discontinuity between conversational AI's recommendation pattern and consumer search query distribution.

The narrowing of v0.12's claim's scope is not a refutation of v0.12's substantive analysis; it is a sharper specification of which categories the v0.12 paper actually describes. The v0.14 results would have been impossible to obtain without v0.12's prior characterisation. The construct-validity arc is incremental: each version sharpens the prior version's scope and surfaces what the prior could not surface.

Limitations

Matched-model subset. AI Presence is measured against a two-model matched subset (Claude Sonnet 4.6 + GPT-5.4-mini). The Tri-System framework's three-mode response taxonomy (Brand mode / Component mode / Authority mode) suggests inter-model variation in AI Presence may be substantial; the matched-subset constraint factors out this variation but does not address it. Future programme phases will report results across additional model families.

Brand-age DRAFT values. 47 of 93 brand-age entries (the v0.14-new entries for skincare and finance) carry DRAFT founding years pending source-URL verification. The H1–H8 results in this report compute on the DRAFT values. The author commits to source-verifying each entry before subsequent programme deposits; founding-year corrections will be applied via a future amendment per DEVIATIONS Entry 3 §3.7.2.

Two-wave short window. v0.14 measures at t_1 (29 April 2026) and t_2 (7 May 2026), separated by eight days. The cross-wave stability that H2 confirms is therefore short-window. The H8 phantom-persistence finding is longer-horizon (Mint's AI Presence is approximately unchanged from a year-prior v0.7 measurement) but is a single-brand observation rather than a population-level longitudinal design.

Five-category panel is still small. Five categories is wider than v0.12's three but small relative to the total category space. The provisional Regime 4 finding rests on two categories (skincare and finance). Future phases will sample additional candidates to test regime stability beyond these two.

Single external validator (Google Trends). The construct-validity test uses Google Trends as the sole consumer-search validator. A multi-validator approach — testing AI Presence simultaneously against search interest, social-media mention rates, retail sales data where available, and consumer-survey aided-recall measures — would generalise the construct-validity claim from a single-validator finding to a multi-validator finding. Such a design is beyond the scope of v0.14 but is the medium-term goal of the AIAS programme.

No incumbent-tier phantoms. v0.14's phantom test is restricted to a single phantom candidate (Mint) in a single category (finance). The Phantom Brand Persistence regularity in v0.7 was observed across multiple challenger-tier brands in cosmetics. v0.14 does not address whether the regularity differs across market tiers or across categories with different operational-closure timing patterns. A multi-phantom design across tiers is a Phase 3 candidate.

What's Next

v0.14 candidate pool. Two cross-lingual candidates remain on the programme's Phase 3 list. Premium tea is a likely Regime 4 candidate (high brand-age dispersion across Latin-script and CJK-script registries). Traditional spirits is a likely Regime 2 or Regime 4 candidate (strong age-mediation expected, with cross-region Trends asymmetry possible). Whether premium tea sits in Regime 3 (descriptive-only via n-floor) or Regime 4 (covariate-saturated weak) is empirically open until measured.

AIAS methodology paper. A short methodology note formalising Regime 4 with threshold-precise definitions analogous to those for Regimes 1–3 is queued behind the Tri-System Brand Growth paper. The AIAS Presence Measurement Protocol v1.1 will incrementally version to v1.2 to incorporate the regime taxonomy as a category-classification step in the canonical measurement pipeline, with the four-regime structure replacing the v0.12 three-regime structure as the protocol's reference taxonomy.

Phantom-persistence cross-tier study. Mint's confirmation in v0.14 extends the Phantom Brand Persistence regularity to personal finance — a category with a single phantom candidate at challenger-tier position. A follow-on designed-for-test phase sampling phantom candidates across incumbent / mid-tier / challenger positions in multiple categories would test whether the regularity's magnitude depends on market tier.

Phase 3 (AIAS components 2–6). Construct validity established for AI Presence in v0.11–v0.14 is a precondition for measurement work on the five remaining AIAS components: Ranking, Consistency, Coverage, Grounding, Sentiment. The v0.14 four-regime finding suggests that each component will require its own per-category construct-validity profile rather than a uniform cross-category correlation with any single external proxy.

External brand-tracking validation. Phase 3 of the programme will test the AIAS Presence component against external brand-tracking data (Kantar BrandZ, YouGov BrandIndex, brand health tracker panels). The four-regime taxonomy provides a structured set of pre-registered predictions: brands in Regime 1 should show stronger AI Presence × brand-tracking correlation than brands in Regimes 3 or 4. The cross-system validity test will sharpen the construct-validity claim from a Trends-only finding to a multi-validator finding.

Hypothesis Scoring

All thresholds and tests locked at *v0.14-prereg* (commit 1a6294d, 11 May 2026 UTC) prior to any Google Trends acquisition call against the wave windows. Per-category pivot exemption from E1b applied per pre-reg §5.1 (sd = 0 by pivot construction). Categories routing to descriptive-only per §3.6a (n < 10 hard floor) carry no H1–H4 inference.

H	Pre-registered prediction	Result	Status
H1 PM	Spearman $\rho > 0.5$ AND $p_{1t} < 0.05$, both waves	$\rho_{t_1} = 0.506$; $\rho_{t_2} = 0.482$	FALSIFIED
H1 Run	Spearman $\rho > 0.5$ AND $p_{1t} < 0.05$, both waves	$\rho_{t_1} = 0.808$ ($p < 0.001$); $\rho_{t_2} = 0.786$ ($p = 0.001$)	**CONFIRMED**
H1 Skin	Spearman $\rho > 0.5$ AND $p_{1t} < 0.05$, both waves	$\rho_{t_1} = 0.282$; $\rho_{t_2} = 0.332$	FALSIFIED
H1 Fin	Spearman $\rho > 0.5$ AND $p_{1t} < 0.05$, both waves	$\rho_{t_1} = 0.168$; $\rho_{t_2} = 0.094$	FALSIFIED
H2 all	$\lvert\Delta\rho\rvert$ [illegible] 0.15 in all four confirmatory categories	PM 0.024; Run 0.022; Skin 0.050; Fin 0.074	**CONFIRMED**
H3 all	AI top-3 Trends top-5, both waves, all confirmatory categories	PM 1/3; Run 2/3; Skin 2/3; Fin 1/3 — falsified in all four	FALSIFIED
H4 PM	Partial $\rho > 0.5$ (age + tier), both waves	partial $\rho = 0.417 / 0.434$	FALSIFIED
H4 Run	Partial $\rho > 0.5$ (age + tier), both waves	partial $\rho = 0.466 / 0.488$	FALSIFIED
H4 Skin	Partial $\rho > 0.5$ (age + tier), both waves	partial $\rho = -0.205 / -0.117$ (covariate-saturated)	FALSIFIED
H4 Fin	Partial $\rho > 0.5$ (age + tier), both waves	partial $\rho = -0.159 / -0.287$ (covariate-saturated)	FALSIFIED
H1–H4 Oil	n-floor [illegible] 10 per wave	n = 8 (WW) below hard floor; carry-forward from v0.12	*Descriptive-only per §3.6a*
H5	v0.12 marginal signature in 3+ of effective-N categories	1 of 4 (PM only; signature absent in running, skincare, finance)	FALSIFIED
H6	Linear-style + Todoist-style co-presence in 4+ of effective-N categories	1 of 4 (PM only; running missing Todoist-style; skincare and finance missing Linear-style)	FALSIFIED
H7	Every category classifies cleanly into one pre-registered regime	3 of 5 clean (PM → R1; Run → R2 boundary; Oil → R3 descriptive); Skin and Fin unclassifiable → provisional Regime 4	FALSIFIED (productive)
H8	Mint phantom signature: AI [illegible] 5% both waves AND AI top-5 both waves AND not Trends top-5 either wave	AI 44.79% / 41.67%; rank 5 / 5; Trends 0.0 both regions both waves (C3 trivially satisfied per §11)	**CONFIRMED**

Hypothesis Details

Per-hypothesis claim, operationalisation, and result. The both-waves conjunction for H1 / H4 provides built-in family-wise error rate control (joint $p \approx 0.05^2 = 0.0025$ under the null). H7 uses the strict all-categories-clean rule for taxonomy under-coverage detection.

PM software replication (Regime 1, Marginal direct). v0.11 / v0.12 construct correlation reproduces at v0.14. Bivariate = 0.506 / 0.482 (within sampling error of v0.11 / v0.12 values). $|\Delta| = 0.024$. H3 falsifies at 1 of 3 both waves with Notion / Jira as the overlapping brands. Partial = 0.417 / 0.434 — decrement of 0.089 / 0.048 from bivariate, well below the Regime 2 threshold. PM software's structural property is now reproducible across three consecutive measurement versions.

Premium running shoes (Regime 2, Age-mediated strong). H1 confirms decisively (bivariate ≈ 0.80 both waves, $p <$ 0.002). H2 confirms $|\Delta| = 0.022$. H3 falsifies at 2 of 3 both waves (Brooks AI-top-3 at t_1, Trends rank 6; Hoka similar at

t_2). H4 falsifies with partial ≈ 0.47 — a decrement of 0.34 from bivariate. Boundary-flagged at t_2 decrement (0.298 within 0.05 of the 0.25 Regime 2 lower bound). The residual partial correlation matches v0.12 PM software's once age + tier is removed.

Premium facial skincare (Regime 4 boundary, lower edge of Regime 1). Bivariate = 0.282 / 0.332 — below 0.35 at t_1, within 0.05 of 0.35 at t_2 (boundary-flagged). H3 falsifies at 2 of 3 both waves. Partial goes ***negative***: −0.205 / −0.117 — covariates absorb the entire bivariate signal and then some. The brand registry is the largest in the v0.14 panel (n = 31 with 28 eligible); the weakness of the rank alignment is not an n-floor artefact.

Personal finance apps (Regime 4 unambiguous). Bivariate = 0.168 / 0.094 — well below 0.35 at both waves. H3 falsifies at 1 of 3 both waves. Partial = −0.159 / −0.287 — unambiguously negative-residual. n = 13 of 15 live brands (Quicken Simplifi E1a-excluded; Mint and Lunch Money Worldwide non-eligible). The category combines genuinely weak rank alignment with the smallest confirmatory-arm n in v0.14.

Premium olive oil (Regime 3, Scale-mismatch). Carry-forward from v0.12. n = 8 Worldwide and n = 7 US, both below the hard floor of 10. Routes to descriptive-only per pre-reg §3.6a. The v0.12 Category-Scale Mismatch finding (7 of 15 matched-subset brands with AI Presence ≥ 5% and Trends below display threshold) carries forward. v0.14 adds no new evidence on olive oil.

H5 cross-category v0.12 marginal-signature replication. Signature definition: in [0.35, 0.65] both waves AND H3 falsified at 1 or 2 of 3 both waves AND ≥ 1 Linear-style brand surfaces. Required: 3+ of effective-N. Applicable categories: PM, Running, Skincare, Finance (Oil descriptive). **FALSIFIED at 1 of 4.** Only PM software matches the signature. Running fails on magnitude (above 0.65). Skincare and finance fail on absence of Linear-style brands. v0.12's marginal signature is PM-software-specific.

H6 cross-category Linear / Todoist co-presence. Linear-style brand (AI ≥ 50% AND Trends ≥ 5) AND Todoist-style brand (AI ≥ 5% AND Trends ≥ 20) both surface at both waves, in 4+ of effective-N categories. **FALSIFIED at 1 of 4.** Only PM software confirms. Running has Linear-style (Brooks / Hoka) but no Todoist-style. Skincare has Todoist-style (Clinique, Olay, Beauty of Joseon) but no Linear-style. Finance has Todoist-style (Origin, Cleo) but no Linear-style. The bidirectional brand-co-presence pattern is PM-software-specific.

H7 three-regimes clean classification. Every category classifies cleanly into exactly one of the three pre-registered regimes (Marginal direct / Age-mediated strong / Scale-mismatch), with no boundary flag (within 0.05 of any threshold) and no unclassifiable-position label. **FALSIFIED at 3 of 5 clean classifications.** PM → R1 clean; Run → R2 boundary (decrement at lower edge); Oil → R3 clean (descriptive route); Skin → unclassifiable (boundary-flagged at Regime 1 lower edge AND partial negative); Fin → unclassifiable (below Regime 1 lower bound at both waves AND partial negative). The strict all-clean rule was chosen for taxonomy under-coverage detection; under-coverage detected. Provisional fourth regime named in this report.

H8 Mint phantom-persistence diagnostic. Three pre-registered conditions: (C1) AI Presence ≥ 5% both waves; (C2) AI Presence top-5 both waves; (C3) NOT Trends top-5 either wave. All three confirmed: AI 44.79% / 41.67% (C1); rank 5 / 5 (C2); Trends 0.0 in both regions at both waves — trivially satisfied per pre-reg §11 since Mint is E1b-ineligible (C3). **CONFIRMED canonically.** The cleanest phantom-persistence anchor in the AIAS programme to date.

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

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Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.1). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

Gonzalez Castro, P. U. (2026). ***Four Empirical Regimes in AI-Mediated Brand Visibility: A Five-Category Replication of the v0.13 Regime 4 (Covariate-saturated Weak) Finding in a Designed-for-Test Category — Premium Tea — AI Presence Index v0.14***. Third System. thirdsystem.ai/v14-premium-tea-regime4-replication (SSRN forthcoming)

Underlying datasets

OSF project ec6wh, /v13/. Inputs (v0.9 AI Presence rates carry-forward for all five categories), Google Trends raw responses (pivot-bundle JSONs across both regions for two waves), Phase B topic-ID resolution logs, pre-acquisition validation outputs (Phase A pivot stability, Phase B solo / bundled-E5 / disambiguation passes), brand age source table (93 rows across 5 categories — 46 v0.12 carry-forward; 47 v0.14-new with DRAFT founding years pending source-URL verification per DEVIATIONS Entry 3 §3.7.2), updated registry files (brands_skincare.json, brands_finance.json added), scoring outputs (canonical_scoring.json, per_brand_paired.csv), build scripts, this report, and the matching SSRN working paper.

Companion SSRN working paper (Gonzalez Castro 2026, SSRN 6750498). Cross-references: AI Availability foundational paper (SSRN 6659000); AIAS Presence Measurement Protocol v1.1 (SSRN 6722319); v0.6 Cross-Category Findings (SSRN 6720959); v0.7 Phantom-Brand Persistence Phase 2 BBB (SSRN 6721779); v0.8 Discourse-Language Knives (SSRN 6728000); v0.9 Longitudinal Re-Baseline (SSRN 6736878); v0.10 Naive-Phantom Rate Stability (SSRN 6741163); v0.11 PM Software × Trends Construct Validity Pilot (SSRN 6745040); v0.12 Three Empirical Regimes (SSRN 6748341).

Methodology log: v0.14 follows AIAS Presence Measurement Protocol v1.1 (unchanged from v0.9 through v0.12). Pre-registration locked at git tag v0.14-prereg (commit 1a6294d) on 11 May 2026 UTC prior to acquisition. Acquisition session UTC timestamp 2026-05-11T21:45:28Z. DEVIATIONS.md Entries 1 and 2 document pre-lock query disambiguation and the Quicken Simplifi E1a exclusion as non-design-altering amendments; Entry 3 documents the empirical Regime 4 finding as a productive falsification of H7 — no pre-registered hypothesis, threshold, or routing rule was modified..

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